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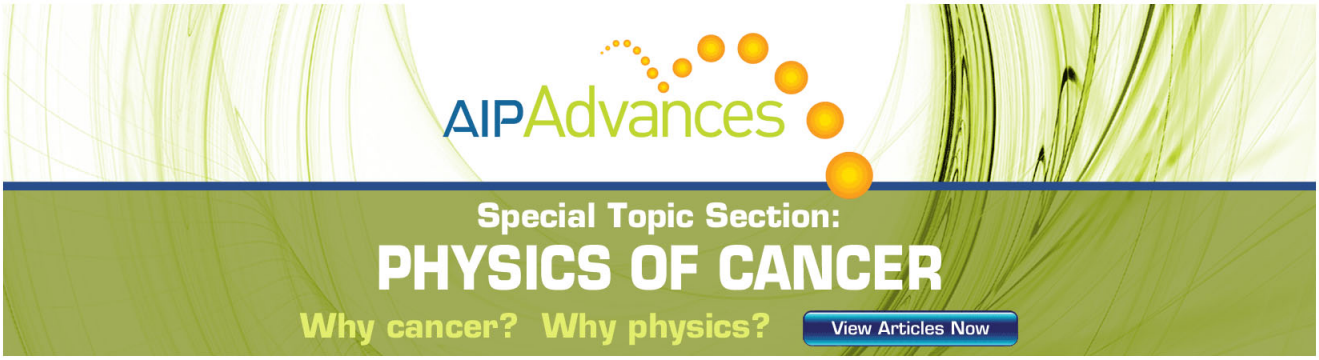

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Evaluation of the grand-canonical partition function using expanded Wang-Landau simulations. I. Thermodynamic properties in the bulk and at the liquid-vapor phase boundary

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The Wang-Landau sampling is a powerful method that allows for a direct determination of the density of states. However, applications to the calculation of the thermodynamic properties of realistic fluids have been limited so far. By combining the Wang-Landau method with expanded grand-canonical simulations, we obtain a high-accuracy estimate for the grand-canonical partition function for atomic and molecular fluids. Then, using the formalism of statistical thermodynamics, we are able to calculate the thermodynamic properties of these systems, for a wide range of conditions spanning the single-phase regions as well as the vapor-liquid phase boundary. Excellent agreement with prior simulation work and with the available experimental data is obtained for argon and CO_2, thereby establishing the accuracy of the method for the calculation of thermodynamic properties such as free energies and entropies. © *2012 American Institute of Physics*. [http://dx.doi.org/10.1063/1.4712023]

I. INTRODUCTION

A few years ago, Wang and Landau proposed a particularly promising method.[1,2] The purpose of the Wang-Landau (WL) approach is different from conventional simulation methods. Instead of calculating equilibrium properties for a given set of thermodynamic conditions, the WL method determines directly the density of states of the system. This approach consists in dynamically updating the density of states (i.e., the degeneracy of energy levels) every time the simulation visits a configuration of the system of a given energy. Then, using the formalism of statistical thermodynamics, all thermodynamic properties of the system can be calculated over a wide range of conditions. The WL method is very appealing since a single simulation run yields the thermodynamic properties of the system for a wide range of conditions. Moreover, unlike conventional simulations, the WL scheme ensures that all energies available to the system are sampled evenly, alleviating the need for additional enhanced sampling techniques. Finally, we add that other methods, such as multicanonical simulations,[3,4] can also give an estimate for the density of states. However, as pointed out by Shell *et al.*,[5] the WL method has the advantage of providing a density of states which can be readily used to calculate the thermodynamic properties, given the direct connection of the WL density of states to the entropy of the system.

In recent years, the WL method was extended to off-lattice simulations,[5,6] generalized to different statistical ensembles[7,8] and applied to study a variety of phenomena,[7,9] including the determination of the conditions for phase coexistence.[8,10–16] Despite its conceptual advantages and its simplicity of use, applications of the WL approach have been limited, partly because of its perceived lack of precision when compared to other methods such as the transition matrix (TM) method.[17–21] For instance, Ganzenmüller and Camp[10] used the WL method in the grand-canonical ensemble to study the vapor-liquid equilibria of the Lennard-Jones fluid. While they reported densities for the two coexisting phases in good agreement with the TM results, they also obtained inaccurate results for the activity and the chemical potential (most notably at low temperatures for which the deviation from the TM results reached 10%). As noted by Ganzenmüller and Camp, the error was intrinsic to the simulation method since it could not be reduced by varying the convergence parameters used in the WL calculation. This is a significant drawback since low-quality estimates for the chemical potential, and hence for the partition function of the system, precludes the direct calculation of the thermodynamic properties for the simulated system.

In this work, we show how we use the WL method to obtain a high-accuracy estimate for the grand-canonical function of atomic and molecular fluids. More specifically, we discuss in Sec. II how we carry out a WL sampling of a simplified version of the expanded grand-canonical ensemble,[22–30] which consists in dividing the insertion and deletion of entire atoms and molecules in a large number of intermediate stages. We discuss the Metropolis criteria obtained for the resulting simulation method. We then apply this method to determine a large number of thermodynamic properties, including the entropy, which is notoriously difficult to determine, for argon and carbon dioxide in the bulk (liquid or vapor) phases as well as at the vapor-liquid phase boundary. We establish the accuracy of the method by comparing our results to prior simulation work, when available, and to experimental data, most notably for the entropy of the bulk phases and at coexistence. We finally draw the main conclusions from this work.

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II. OBTAINING A HIGH-ACCURACY ESTIMATE FOR THE GRAND-CANONICAL PARTITION FUNCTION

A. Formalism

The grand-canonical partition function for a system of N atoms can be written as

$$\Theta(\mu, V, T) = \sum_{N=0}^{\infty} Q(N, V, T) \exp(\beta\mu N), \tag{1}$$

where $\beta = 1/k_B T$, μ is the chemical potential, and $Q(N, V, T)$ is the canonical partition function given by

$$Q(N, V, T) = \frac{V^N}{N!\Lambda^{3N}} \int \exp(-\beta U(\mathbf{\Gamma}))d\mathbf{\Gamma}, \tag{2}$$

where the integration is performed over the coordinates of the N atoms of the system ($\mathbf{\Gamma}$ denotes a specific configuration of the system of N atoms) and Λ is the De Broglie wavelength.

Accurately sampling the grand-canonical ensemble requires efficient schemes to insert additional particles into the system (and to delete existing particles). Since the insertion and deletion of entire atoms and molecules are often problematic, it has been suggested to divide the insertion, or deletion, in M stages, where M is an integer. At all times, the system is therefore composed of N full particles and of a fractional particle, which, for a given stage l ($0 \le l \le M-1$), interacts with the full particles through a coupling parameter ξ_l. l is allowed to change and can be interpreted as a fractional number of particle. If $l = 0$, the system consists of N full particles and there is no fractional particle. The partition function for this expanded grand-canonical (EGC) (Refs. 22 and 24–30) ensemble of N full particles and a fractional particle is given by

$$\Theta_{\mathrm{EGC}}(\mu, V, T) = \sum_{N=0}^{\infty} \sum_{l=0}^{M-1} Q(N, V, T, l) \exp(\beta\mu N + \psi(l, N)), \tag{3}$$

where $\psi(l, N)$ is a weighting function. This weighting function $\psi(l, N)$ is generally determined numerically[25,31,32] to optimize the sampling around a given value of N (i.e., the value of N around which conventional grand-canonical or Gibbs Ensemble Monte Carlo simulations[33] converge).

In Eq. (3), $Q(N, V, T, l)$ is the canonical partition function for a system of N full particles and a fractional particle at stage $l > 0$, given by

$$Q(N, V, T, l) = \frac{V^{N+1}}{N!\Lambda^{3N}\Lambda_l^3} \int \exp(-\beta U(\mathbf{\Gamma}))d\mathbf{\Gamma}, \tag{4}$$

where $\mathbf{\Gamma}$ denotes a specific configuration of the system consisting of N particles and of the fractional particle and Λ_l is the De Broglie wavelength for the fractional particle in stage l. For $l = 0$, $Q(N, V, T, 0)$ is the canonical partition function for a system of N particles (Eq. (2)). Therefore, if we choose to impose $\psi(0, N) = 0$, we recover the partition function of the grand-canonical ensemble by considering only the terms for which $l = 0$ in Eq. (3).

The aim of this work is to evaluate the grand-canonical partition function. Recognizing that the choice of the weighting function $\psi(l, N)$ does not affect the value of the grand-canonical partition function, we first simplify Eq. (3) by imposing $\psi(l, N) = 0$ for all l and all N. We then choose to impose that for all l, the mass of the fractional particle is the same as that of the full particle. This implies that $\Lambda_l = \Lambda$. We obtain a simplified expanded grand-canonical (SEGC) ensemble with the following partition function:

$$\Theta_{\mathrm{SEGC}}(\mu, V, T) = \sum_{N=0}^{\infty} \sum_{l=0}^{M-1} Q(N, V, T, l) \exp(\beta\mu N) \tag{5}$$

with, for $0 < l < M$,

$$Q(N, V, T, l) = \frac{V^{N+1}}{N!\Lambda^{3(N+1)}} \int \exp(-\beta U(\mathbf{\Gamma}))d\mathbf{\Gamma}. \tag{6}$$

B. Determination of the grand-canonical partition function using a Wang-Landau sampling

We now propose to use the Wang-Landau sampling method to evaluate the SEGC partition function $\Theta_{\mathrm{SEGC}}(\mu, V, T)$. In the simplified expanded grand-canonical ensemble, the joint Boltzmann distribution for Γ, N, and l is

$$\begin{aligned} p(\mathbf{\Gamma}, N, l) &= \frac{V^{N+1}\exp(\beta\mu N - \beta U(\mathbf{\Gamma}))}{N!\Lambda^{3(N+1)}\Theta_{\mathrm{SEGC}}(\mu, V, T)} \text{ for } 0 < l < M, \\ &= \frac{V^{N}\exp(\beta\mu N - \beta U(\mathbf{\Gamma}))}{N!\Lambda^{3N}\Theta_{\mathrm{SEGC}}(\mu, V, T)} \quad \text{for } l = 0. \end{aligned} \tag{7}$$

The distribution $p(N, l)$ can then be estimated as

$$p(N, l) = \int p(\mathbf{\Gamma}, N, l)d\mathbf{\Gamma} = \frac{Q(N, V, T, l)\exp(\beta\mu N)}{\Theta_{\mathrm{SEGC}}(\mu, V, T)}. \tag{8}$$

We finally define the biased distribution $p_{bias}(\mathbf{\Gamma}, N, l) = p(\mathbf{\Gamma}, N, l)/p(N, l)$ to obtain uniform sampling of (N, l) within the Wang-Landau scheme. It follows that the Metropolis criterion for a move from an old state (state point $\mathbf{\Gamma}_o$; number of full particles N_o and fractional number l_o) to a new state ($\mathbf{\Gamma}_n$; N_n and l_n) is given by[10]

$$acc(o \to n) = min\left[1, \frac{p_{bias}(\mathbf{\Gamma}_n, N_n, l_n)}{p_{bias}(\mathbf{\Gamma}_o, N_o, l_o)}\right]. \tag{9}$$

More specifically, if l_o and l_n are both either equal to 0 or both strictly larger than 0, the Metropolis criterion becomes

$$acc(o \to n) = min\left[1, \frac{Q(N_o, V, T, l_o)V^{N_n}N_o!\Lambda^{3N_o}\exp(-\beta U(\mathbf{\Gamma}_n))}{Q(N_n, V, T, l_n)V^{N_o}N_n!\Lambda^{3N_n}\exp(-\beta U(\mathbf{\Gamma}_o))}\right]. \tag{10}$$

If $l_o = 0$ and $l_n > 0$, the Metropolis criterion becomes

$$acc(o \to n) = min\left[1, \frac{Q(N_o, V, T, l_o)V^{N_n+1}N_o!\Lambda^{3N_o}\exp(-\beta U(\mathbf{\Gamma}_n))}{Q(N_n, V, T, l_n)V^{N_o}N_n!\Lambda^{3(N_n+1)}\exp(-\beta U(\mathbf{\Gamma}_o))}\right] \tag{11}$$

and if $l_o > 0$ and $l_n = 0$, the Metropolis criterion becomes

$$acc(o \to n) = min\left[1, \frac{Q(N_o, V, T, l_o)V^{N_n}N_o!\Lambda^{3(N_o+1)}\exp(-\beta U(\mathbf{\Gamma}_n))}{Q(N_n, V, T, l_n)V^{N_o+1}N_n!\Lambda^{3N_n}\exp(-\beta U(\mathbf{\Gamma}_o))}\right]. \tag{12}$$

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This formalism can be extended to molecular fluids by including q_{int}, the molecular partition function for the internal degrees of freedom of rotation (e.g., in this work since we model CO_2 as a rigid molecule) and vibration. The Metropolis criterion becomes for a molecular fluid

$$acc(o \rightarrow n) = min\left[1, \frac{Q(N_o, V, T, l_o) V^{N_n} q_{int}^{N_n} N_o! \Lambda^{3N_o} \exp(-\beta U(\mathbf{\Gamma}_n))}{Q(N_n, V, T, l_n) V^{N_o} q_{int}^{N_o} N_n! \Lambda^{3N_n} \exp(-\beta U(\mathbf{\Gamma}_o))}\right]. \quad (13)$$

As can be seen from the equations above, the chemical potential does not appear in the Metropolis criteria. This means that we do not need to specify a given value for μ to run the simulation. The Wang-Landau sampling scheme leads to a self-consistent determination of $Q(N, V, T, l)$. At the beginning of the simulation, we impose $Q(N, V, T, l) = 1$ for all N and l. During the simulation, whenever the system visits a configuration with N full particles and a fractional particle at stage l, we update the value of $Q(N, V, T, l)$ (i.e., by multiplying it by f, a convergence factor greater than 1). Throughout the simulation, we collect a histogram of the number of visits for each N and l. Once this histogram is close enough to being flat, the convergence factor is reduced, the histogram for the number of visit is initialized and the simulation is run again with this new convergence factor. We add that the convergence of the Wang-Landau algorithm has been formally proved[34] and that there is a considerable freedom in choosing a criterion for estimating whether the histogram $H(V)$ is flat. This process is repeated until f reaches a minimum threshold value, leading to an accurate numerical estimate for $Q(N, V, T, l)$ for all N and l. Since the Wang-Landau algorithm provides $Q(N, V, T, l)$ up to a multiplicative factor, we add that the actual value for $Q(N, V, T, l)$ is determined by recognizing that $Q(N = 0, V, T, l = 0)$ is equal to 1. Finally, we obtain the numerical estimate for the grand-canonical partition function as

$$\Theta(\mu, V, T) = \sum_{N=0}^{\infty} Q(N, V, T, l = 0) \exp(\beta \mu N). \quad (14)$$

III. SIMULATION MODELS AND TECHNIQUES

A. Argon

The first system studied in this work is argon, modeled with the Lennard-Jones (LJ) fluid. The interactions between two LJ particles i and j are given by

$$\phi(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right], \quad (15)$$

where r_{ij} is the distance between the two particles i and j, σ is the exclusion diameter, and ε is the depth of the potential well.

The fractional LJ particle i interacts with j, one of the N full particles through the following equation:

$$\phi(r_{ij}) = 4\epsilon_{\xi_l} \left[\left(\frac{\sigma_{\xi_l}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{\xi_l}}{r_{ij}}\right)^{6}\right], \quad (16)$$

where ξ_l denotes the coupling parameter, r_{ij} represents the distance between i and j, $\epsilon_{\xi_l} = (l/M)^{1/3}\epsilon$ and $\sigma_{\xi_l} = (l/M)^{1/4}\sigma$ ($M = 100$ is the number of stages for the insertion/deletion of the fractional particle and l varies between 0 and $M - 1$). The scaling exponents for ϵ and σ are taken from previous work.[35] The Lennard-Jones parameter for argon are $\varepsilon/k_B = 117.05$ K and $\sigma = 3.4$ Å. In line with prior work,[10,17] most of the results presented here were generated for a volume of $V = 512\sigma^3$ and a cutoff distance for the calculations of pair interactions to 3σ. We set the maximum number of argon atoms to 450, which allows us to study the vapor and the liquid regions of the phase diagram of argon, as well as the vapor-liquid coexistence of argon. In addition, in calculations aiming at determining the dependence of the surface tension on the system size, we carried out simulations with volumes of $V = 729\sigma^3$, $V = 1000\sigma^3$, and $V = 1728\sigma^3$. Error estimates were calculated as the standard deviation for each property over a set of four independent expanded Wang-Landau simulations.

B. Carbon dioxide

The second system we study is the CO_2 molecule. In this work, CO_2 molecules interact with each other through the TraPPE potential,[36] which consists of 3 LJ sites and 3 point charges per molecule (i.e., 1 LJ site and 1 charge per atom). The interaction between two atoms i and j (either C or O), belonging to two different CO_2 molecules, is given by

$$\phi(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}. \quad (17)$$

The parameters for the model[36] are as follows: $\epsilon_{CC}/k_B = 27$ K, $\epsilon_{OO}/k_B = 79$ K, $\sigma_{CC} = 2.80$ Å, $\sigma_{OO} = 3.05$ Å, $q_C = 0.7e$, and $q_O = -0.35e$. The parameters for the LJ interactions between unlike atoms are calculated using the Lorentz-Berthelot mixing rules.[37] The interaction of an atom i belonging to the fractional CO_2 molecule with an atom j belonging to one of the N full CO_2 molecules is calculated according to

$$\phi(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij,\xi_l}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij,\xi_l}}{r_{ij}}\right)^{6}\right] + \frac{q_{i,\xi_l} q_j}{4\pi\epsilon_0 r_{ij}}, \quad (18)$$

where ξ_l denotes the coupling parameter, r_{ij} represents the distance between atoms i and j, $\epsilon_{ij,\xi_l} = (l/M)^{1/3}\epsilon_{ij}$, $\sigma_{ij,\xi_l} = (l/M)^{1/4}\sigma_{ij}$, and $q_{i,\xi_l} = (l/M)^{1/3} q_i$ (in line with argon, $M = 100$ is the number of stages for the insertion/deletion of the fractional particle and l varies between 0 and $M - 1$). In addition to using a partial coupling for the interactions, we also apply a homothetic transformation to the skeleton of the fractional molecule as the fractional molecule is shrunk or grown. This implies that, for a fractional molecule at a given stage l, the distance between the carbon and one of the oxygens is equal to $(l/M)^{1/4} d_{C-O}^{eq}$, where $d_{C-O}^{eq} = 1.16$ Å is the distance between C and O in a full molecule of carbon dioxide. Most of the results presented here were obtained for a boxlength of $L = 31.05$ Å with a cutoff distance for the

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calculations of pair interactions set to half the boxlength. As we did for the Lennard-Jones system, we also carried out simulations with boxlengths of $L = 35.54$ Å, $L = 39.12$ Å, and $L = 44.78$ Å to determine the dependence of the surface tension on the system size.

C. Simulation details

We combine the Wang-Landau sampling scheme with Monte Carlo simulations. For argon, the Monte Carlo (MC) steps consist in either the translation of a single atom (75% of the MC steps) or in a change in (N, l) (25% of the MC steps). For CO_2, the MC steps are either the translation of a single molecule (37.5% of the MC steps), the rotation of a single molecule (37.5% of the MC steps) or in a change in (N, l) (25% of the MC steps). The initial value for the convergence factor is set to e. After every iteration of the process, the value of the convergence factor is reduced by taking its square root until the minimal threshold value ($f = 10^{-8}$) is reached. As noted by Ganzenüller and Camp,[10] the convergence of the Wang-Landau algorithm has been formally proved[34] and there is a considerable freedom in choosing a criterion for estimating whether the histogram $H(V)$ is flat. In this work, we use the same criterion as in Ref. 10 (i.e., that all values of N and l have been visited at least 1000 times) to determine that the Wang-Landau simulation has converged for a given value of the convergence factor. The simplicity of this method lies in the fact that we do not need to specify any value for the chemical potential to run the simulation. Moreover, we only need to run a single simulation to have access to the partition function at a given temperature.

IV. RESULTS AND DISCUSSION

A. Argon

Once $Q(N, V, T, l)$ is obtained, we analyze the results obtained for $Q(N, V, T, l = 0)$. We will note $Q(N, V, T, l = 0)$ as $Q(N, V, T)$ in the rest of the paper. We present in Fig. 1(a) the reduced canonical partition function $Q^*(N, V, T) = \Lambda^{3N} Q(N, V, T)$ as a function of the number of argon atoms for temperatures between 87.79 K and 140.46 K. Once $Q(N, V, T)$ is known, we can evaluate the grand-canonical partition function $\Theta(\mu, V, T)$ through Eq. (14). Equation (14) is an infinite sum over N of $Q(N, V, T)\exp(\beta\mu N)$. This sum converges if, for large values of N, the term $Q(N, V, T)\exp(\beta\mu N)$ tends towards 0. We plot in Fig. 1(b) the variations of $Q^*(N, V, T)\exp(\beta\mu N)$, scaled by its maximum value, for different values of the chemical potential μ. Figures 1(b) and 1(c) show that, for large N, this term becomes negligible and that the numerical estimate for the grand-canonical partition function has converged. From there, using the formalism of statistical mechanics, we can calculate the thermodynamic properties of single-phase (bulk) systems and at phase boundaries. In what follows, we focus on the properties at the vapor-liquid coexistence and in the bulk (liquid or vapor). To assess the accuracy of the grand-canonical partition function, we compare our results to experimental data and available simulation results.

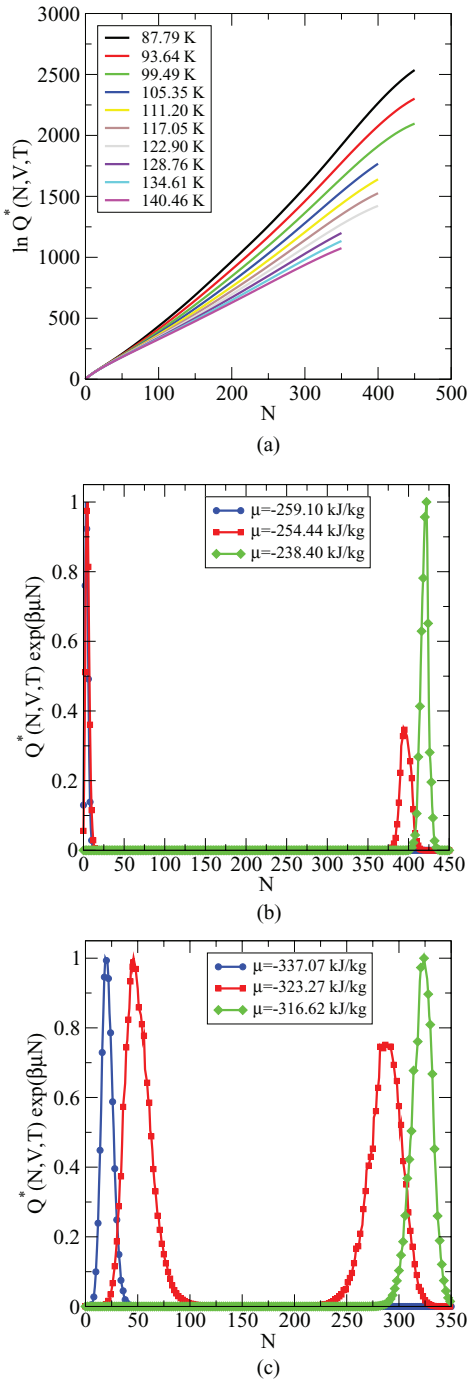

FIG. 1. Argon: (a) Logarithm of the reduced canonical partition function $Q^*(N, V, T) = \Lambda^{3N} Q(N, V, T)$ against the number of atoms. Convergence of the numerical estimate for the grand-canonical partition function as shown by the variation of $Q^*(N, V, T)\exp(\beta\mu N)$, scaled by its maximum value for each chemical potential, as a function of N for (b) $T = 99.49$ K and (c) $T = 140.46$ K.

1. Thermodynamic properties at the liquid-vapor coexistence

We start by explaining how we use the grand-canonical partition function to determine the conditions of coexistence. Our first task is to locate the phase transition. This can be done by determining the variation of the chemical potential with the average number of argon atoms (Fig. 2). For this purpose, we vary the chemical potential and, for each value of

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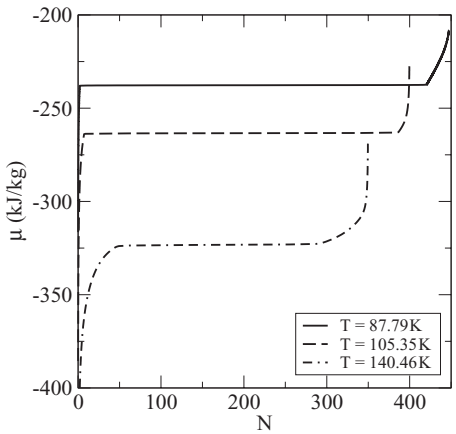

FIG. 2. Argon: Chemical potential μ (in kJ/kg) against the average number of atoms $\langle N \rangle$.

μ, we use $Q(N, V, T)$ to calculate the average number of argon atoms as

$$\langle N \rangle = \frac{\sum_N N \left[Q(N, V, T) \exp(\beta \mu N) \right]}{\Theta(\mu, V, T)}. \tag{19}$$

For each value of T, plotting μ as a function of $\langle N \rangle$ results in a curve exhibiting a plateau (see Fig. 2) at the liquid-vapor coexistence. This provides a rough estimate for μ_{coex}, the chemical potential at coexistence.

Alternatively, μ_{coex} can be located by plotting the variations of the pressure P with μ (Fig. 3). For each value of μ, P can be evaluated from $Q(N, V, T)$ as

$$P = \frac{k_B T \ln \Theta(\mu, V, T)}{V}. \tag{20}$$

Once an estimate for μ_{coex} is obtained, we can further refine its value by analyzing the number distribution $p(N)$, defined as

$$p(N) = \frac{Q(N, V, T) \exp(\beta \mu N)}{\Theta(\mu, V, T)}. \tag{21}$$

Depending on the value of μ, the number distribution exhibits either one peak, corresponding to the bulk, either liquid (Fig. 4(a)) or vapor (Fig. 4(b)), or two peaks, when the system

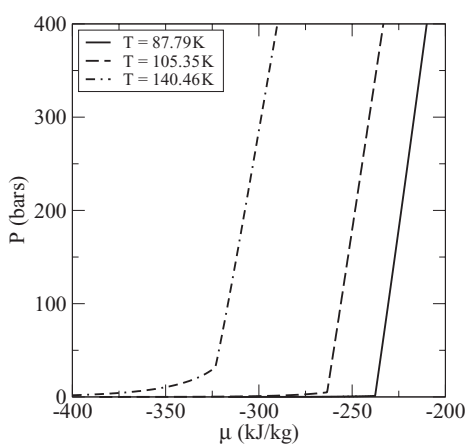

FIG. 3. Argon: Pressure P (in bars) against the chemical potential μ (in kJ/kg).

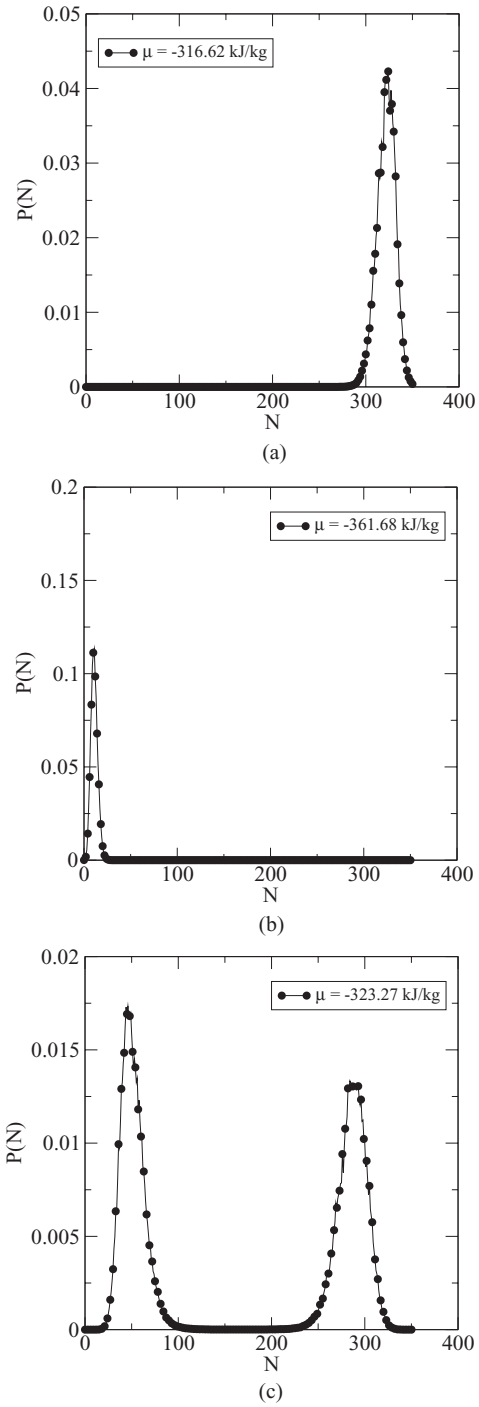

FIG. 4. Argon: Number distribution $p(N)$ at $T = 140.46$ K for different values of μ: (a) $\mu = -316.62$ kJ/kg (bulk liquid), (b) $\mu = -361.68$ kJ/kg (bulk vapor), and (c) $\mu = -323.27$ kJ/kg (liquid-vapor coexistence).

is close to coexistence. The chemical potential at coexistence μ_{coex} corresponds to the value of μ for which the areas below the two peaks are equal to each other (see Fig. 4(c)), indicating that the two phases are equally probable for $\mu = \mu_{coex}$. We provide in Table I the values for μ_{coex} and for the activity at coexistence ($z_{sat} = \Lambda^{-3} \exp(\beta \mu_{coex})$). An excellent agreement is obtained between the results obtained in this work and the results from the TMMC approach.[17] In particular, the method presented in this work accurately predicts the activity at coexistence as shown in Table I. This is a significant improvement

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TABLE I. Argon: Chemical potential μ_{coex} (in kJ/kg) at the vapor-liquid coexistence (standard deviation of 0.07 kJ/kg) and activity z (in reduced units). Transition matrix Monte Carlo results are taken from Ref. 17.

T(K)	μ_{coex}(kJ/kg) this work	$\ln z_{sat}$ this work	$\ln z_{sat}$ TMMC
140.46	−323.27	−3.028	−3.031
134.61	−312.65	−3.193	−3.197
128.76	−302.32	−3.382	−4.384
122.90	−292.16	−3.592	−3.595
117.05	−282.35	−3.833	−3.834
111.20	−272.73	−4.104	−4.107
105.35	−263.41	−4.415	−4.419
99.49	−254.44	−4.775	−4.779
93.64	−245.87	−5.194	−5.196
87.79	−237.68	−5.684	−5.683

over the Wang-Landau approach presented in Ref. 10, which exhibited departures of the order of 10% at low temperatures.

We now turn to the determination of the densities of the two coexisting phases through

$$\langle \rho_{liq} \rangle = \frac{\sum_{N>N_b} \frac{N}{V} Q(N,V,T)\exp(\beta\mu N)}{\sum_{N>N_b} Q(N,V,T)\exp(\beta\mu N)} \tag{22}$$

for the liquid density, where N_b is the number of Ar atoms for which $p(N)$ reaches its minimum, and

$$\langle \rho_{vap} \rangle = \frac{\sum_{N<N_b} \frac{N}{V} Q(N,V,T)\exp(\beta\mu N)}{\sum_{N<N_b} Q(N,V,T)\exp(\beta\mu N)} \tag{23}$$

for the vapor density. We present in Table II the results obtained in this work for the densities at coexistence and compare them to the experimental data and to the TMMC results. Once again, we obtain an excellent agreement between the three sets of results, further establishing the reliability and accuracy of the method presented in this work.

We move on to the determination of the thermodynamic properties at coexistence. We first plot the phase diagram of argon in the pressure-temperature plane (Fig. 5) and in the volume-temperature plane (Fig. 6). In Figs. 5 and 6, we report the critical point estimated from our simulations with the usual scaling laws.[39,40] Both plots show that the agreement

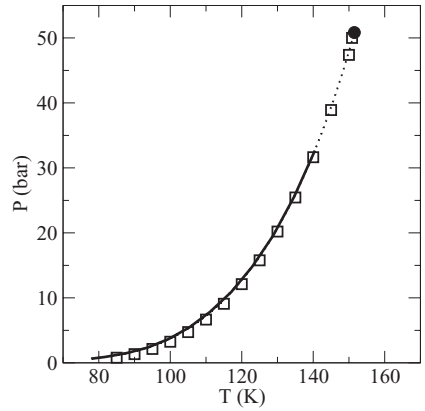

FIG. 5. Argon: Vapor-liquid phase diagram in the pressure-temperature plane. Results obtained in this work are connected by a solid line, while the experimental data are shown with squares. The critical point estimated from the simulations is shown as a filled circle.

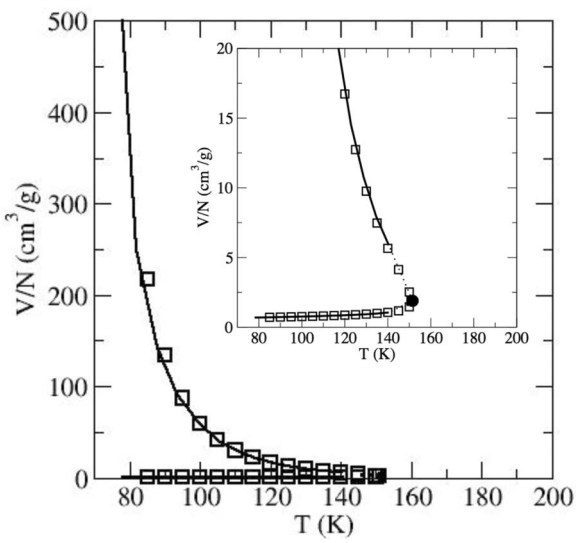

FIG. 6. Argon: Vapor-liquid phase diagram in the volume-temperature plane. Results obtained in this work are connected by a solid line, while the experimental data are shown with squares. The critical point estimated from the simulations is shown as a filled circle.

TABLE II. Argon: Liquid and vapor densities (in g/cm^3) at coexistence (standard deviations of 0.002 g/cm^3 and 0.002 g/cm^3 for the liquid and vapor densities, respectively). Experimental data are taken from Ref. 38, while transition matrix Monte Carlo results are obtained from Ref. 17.

T (K)	ρ_{liq} (g/cm^3) this work	ρ_{liq} (g/cm^3) TMMC	ρ_{liq} (g/cm^3) exp.	ρ_{vap} (g/cm^3) this work	ρ_{vap} (g/cm^3) TMMC	ρ_{vap} (g/cm^3) exp.
140.46	0.947	0.953	0.934	0.170	0.170	0.182
134.61	1.026	1.025	1.012	0.126	0.125	0.131
128.76	1.086	1.085	1.077	0.093	0.093	0.096
122.90	1.139	1.138	1.134	0.069	0.069	0.070
117.05	1.191	1.187	1.184	0.050	0.050	0.051
111.20	1.230	1.232	1.231	0.036	0.036	0.036
105.35	1.275	1.274	1.273	0.025	0.025	0.025
99.49	1.311	1.315	1.314	0.0165	0.0163	0.0163
93.64	1.354	1.354	1.352	0.011	0.010	0.010
87.79	1.389	1.391	1.389	0.007	0.006	0.006

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with the experimental data is very good over the whole temperature range with an estimated critical point with $\rho_c = 0.528 \pm 0.008$ g/cm^3, $T_c = 151 \pm 4$ K, and $P_c = 50.8 \pm 2$ bar close to the experimental critical point[38] at $\rho_c = 0.536$ g/cm^3, $T_c = 150.86$ K, and $P_c = 50.0$ bar. The partition function obtained from the expanded Wang Landau simulations can be further used to determine the value taken by various state functions at coexistence.

We start with the evaluation of the entropies S_{liq} and S_{vap} at coexistence. While the determination of the entropy is notoriously difficult to achieve in simulations, the expanded Wang Landau simulations provide a direct route to evaluate S_{liq} and S_{vap} through

$$S_{liq} = \frac{k_B \ln \Theta(\mu,V,T)}{\langle N_{liq}\rangle} + \frac{(U_{liq}-\mu)}{T},$$
$$S_{vap} = \frac{k_B \ln \Theta(\mu,V,T)}{\langle N_{vap}\rangle} + \frac{(U_{vap}-\mu)}{T} \quad (24)$$

with the internal energy for the liquid U_{liq} and for the vapor U_{vap} at coexistence defined as

$$U_{liq} = \frac{\sum_{N>N_b}\left(E_{pot}(N)+\frac{3}{2}k_BT\right)p(N)}{\sum_{N>N_b} p(N)},$$
$$U_{vap} = \frac{\sum_{N<N_b}\left(E_{pot}(N)+\frac{3}{2}k_BT\right)p(N)}{\sum_{N<N_b} p(N)} \quad (25)$$

in which $E_{pot}(N)$ is the average potential energy per argon atom, collected during the expanded Wang Landau simulation, for a system containing N full argon atoms (and a fractional particle with $l = 0$). In this expression, $\frac{3}{2}k_BT$ is the kinetic (ideal gas) contribution arising from the 3 translational degrees of freedom per argon atom.

We plot in Fig. 7 the entropies of the two coexisting phases in the volume-entropy plane and report the data in Table III, together with the experimental data. Table III demonstrates that we obtain an excellent agreement with the experiment for the liquid (with deviations between 0.01 kJ/kg/K and 0.02 kJ/kg/K over the whole range of temperatures) and a good agreement for the vapor (with deviations between 0.01 kJ/kg/K and 0.05 kJ/kg/K).

The Helmholtz free energy can then be calculated for both the liquid and the vapor phases according to

$$A_{liq} = U_{liq} - TS_{liq},$$

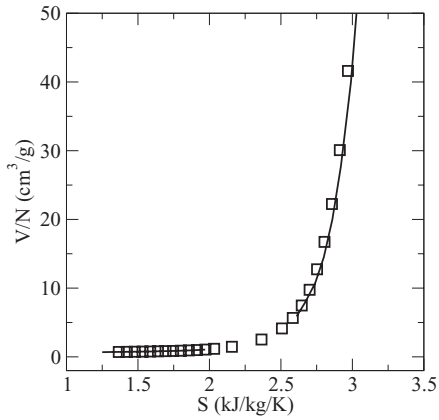

FIG. 7. Argon: Vapor-liquid phase diagram in the volume-entropy plane. Results obtained in this work are connected by a solid line, while the experimental data are shown with squares.

$$A_{vap} = U_{vap} - TS_{vap}. \quad (26)$$

The Helmholtz free energies so obtained are given in Table IV. Table IV also provides a comparison with the corresponding experimental values for A. The agreement between the predicted value and the experimental data for the Helmholtz free energies is remarkably good for both phases (well within 0.5%), resulting in errors of the order of 1 kJ/kg for the Helmholtz free energy change upon vaporization. We finally conclude our examination of the thermodynamic state functions at coexistence by comparing the Gibbs free energy obtained from the expanded Wang-Landau simulations to the experimental data in Table V. Once again, the agreement is very good, i.e., within 0.6% over the whole set of conditions studied in this work.

2. Thermodynamic properties of bulk phases

We now turn to the results obtained for bulk phases. These can be easily obtained from the grand-canonical partition function for a wide range of conditions. This is done by gradually varying the chemical potential with $\mu \leq \mu_{coex}$ to determine the thermodynamic properties for the vapor phase and $\mu \geq \mu_{coex}$ for the liquid phase. In both cases, the number probability distribution exhibits a single peak (see

TABLE III. Entropies at coexistence for argon (standard deviations of 0.002 kJ/kg/K for the liquid and of 0.0008 kJ/kg/K for the vapor). Experimental data are taken from Ref. 38.

T (K)	S_{liq} (kJ/kg/K) this work	S_{liq} (kJ/kg/K) exp.	S_{vap} (kJ/kg/K) this work	S_{vap} (kJ/kg/K) exp.	ΔS(kJ/kg/K) this work	ΔS(kJ/kg/K) exp.
140.46	1.97	1.98	2.61	2.58	0.64	0.60
134.61	1.90	1.91	2.67	2.65	0.78	0.74
128.76	1.84	1.84	2.74	2.73	0.90	0.89
122.90	1.77	1.78	2.80	2.78	1.02	1.00
117.05	1.71	1.71	2.86	2.84	1.15	1.12
111.20	1.65	1.65	2.92	2.91	1.27	1.26
105.35	1.59	1.59	2.99	2.97	1.41	1.38
99.49	1.53	1.53	3.07	3.04	1.55	1.51
93.64	1.45	1.46	3.16	3.12	1.71	1.66
87.79	1.38	1.40	3.27	3.20	1.88	1.92

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TABLE IV. Helmholtz free energies at coexistence for argon. Experimental data are taken from Ref. 38.

T(K)	A_{liq} (kJ/kg) this work	A_{liq} (kJ/kg) exp.	A_{vap} (kJ/kg) this work	A_{vap} (kJ/kg) exp.	ΔA(kJ/kg) this work	ΔA(kJ/kg) exp.
140.46	−326.26	−327.83	−342.23	−342.18	−15.97	−14.35
134.61	−315.36	−316.15	−332.29	−332.54	−16.93	−16.39
128.76	−304.53	−304.54	−323.92	−324.94	−19.39	−20.40
122.90	−292.96	−294.45	−314.20	−313.75	−21.24	−19.30
117.05	−283.22	−283.26	−304.46	−303.69	−21.24	−20.43
111.20	−272.88	−273.72	−294.40	−294.85	−21.52	−21.13
105.35	−264.13	−264.47	−285.00	−284.34	−20.87	−19.87
99.49	−255.14	−255.72	−275.70	−274.45	−20.56	−18.73
93.64	−245.62	−246.45	−265.44	−265.03	−19.82	−18.58
87.79	−237.39	−238.59	−254.70	−254.87	−17.31	−16.28

Figs. 4(a) and 4(b)). The equations giving the density and the thermodynamic functions become

$$\begin{aligned}\langle\rho\rangle &= \frac{\sum_N \frac{N}{V} Q(N,V,T)\exp(\beta\mu N)}{\sum_N Q(N,V,T)\exp(\beta\mu N)},\\ U &= \frac{\sum_N \left(E_{pot}(N)+\frac{3}{2}k_BT\right)p(N)}{\sum_N p(N)},\\ S &= \frac{k_B \ln\Theta(\mu,V,T)}{\langle N\rangle} + \frac{(U-\mu)}{T},\\ A &= U - TS. \end{aligned} \tag{27}$$

We show in Fig. 8 the variation of density (calculated using the first line of Eq. (27)) with pressure in the single-phase vapor region of the phase diagram (Fig. 8(a)) and in the single-phase liquid region of the phase diagram (Fig. 8(b)) for $T = 105.35$ K and $T = 140.46$ K. Both plots show that the density-pressure relationship obtained from the grand-canonical partition function is in excellent agreement with the experimental data, further establishing the high-quality of the results from the expanded Wang-Landau simulations. We then use the third line of Eq. (27) to evaluate the entropy as a function of pressure for the bulk vapor and liquid phases. The results are shown in Fig. 9, together with a comparison with the entropies determined experimentally. Once again, the plots show a very good agreement between the calculated entropies and the experimental data, which show the reliability for bulk phases of the approach presented here.

3. Surface tension calculations

The surface tension is an example of yet another property that can be determined from the simulations presented in this work. Starting from the number probability distribution $p(N)$ at coexistence, defined in Eq. (21), we determine for a system of finite dimension L, where L is the edge of the cubic box in the simulations, the L-dependent interfacial free energy[41] as

$$F_L = \frac{1}{2}\left(\ln p_l^{max} + \ln p_v^{max}\right) - \ln p(N_b). \tag{28}$$

In Eq. (28), p_l^{max} is the maximum probability for the peak of $p(N)$ corresponding to the liquid (see, e.g., on the right of Fig. 4(c)), p_v^{max} is the maximum probability for the peak

TABLE V. Gibbs free energy for argon at coexistence. Experimental data are taken from Ref. 38.

T(K)	140.46	134.61	128.76	122.90	117.05
G(kJ/kg) (this work)	− 323.27	− 312.65	− 302.32	− 292.16	− 282.35
G(kJ/kg) (exp)	− 324.39	− 313.56	− 303.95	− 293.37	− 282.91
T (K)	111.20	105.35	99.49	93.64	87.79
G(kJ/kg) (this work)	− 272.73	− 263.41	− 254.44	− 245.87	− 237.68
G(kJ/kg) (exp)	− 273.96	− 264.35	− 255.36	− 246.44	− 237.86

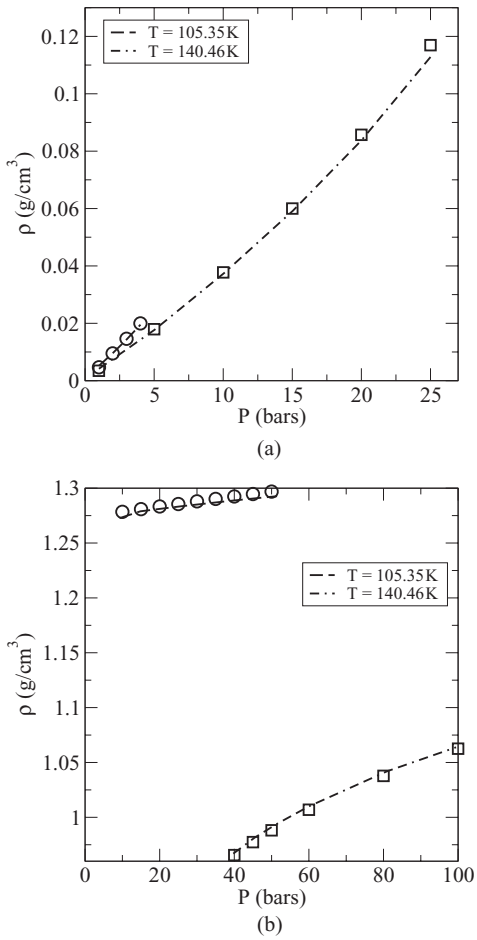

FIG. 8. Single-phase bulk properties for argon: Variations of density as a function of pressure for (a) the vapor and (b) the liquid. Results obtained in this work are shown with lines, while the experimental data are shown with circles (105.35 K) and squares (140.46 K).

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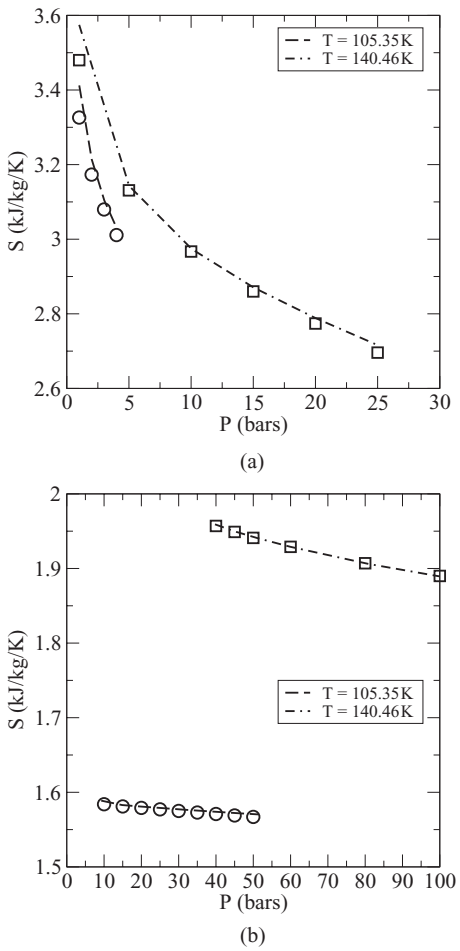

FIG. 9. Single-phase bulk properties for argon: Variations of entropy as a function of pressure for (a) the vapor and (b) the liquid. Results obtained in this work are shown with lines, while the experimental data are shown with circles (105.35 K) and squares (140.46 K).

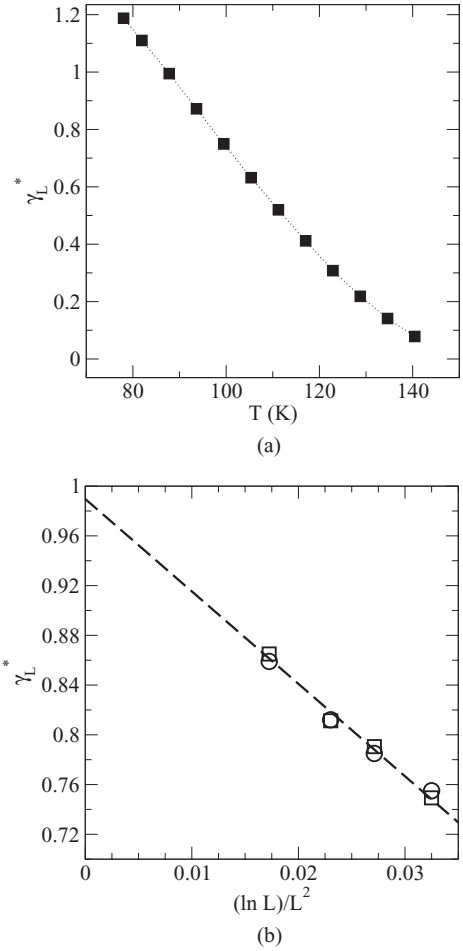

FIG. 10. Surface tension for argon: Variations of surface tension (in units reduced with respect to the Lennard-Jones parameters) as a function of (a) temperature and (b) system size for a reduced temperature ($T^* = k_BT/\epsilon$) of 0.85. Results obtained in this work are shown as squares, while the TMMC results are shown as circles.

in $p(N)$ corresponding to the vapor (see, e.g., on the left of Fig. 4(c)) and $p(N_b)$ is the minimum for the probability, reached for $N = N_b$.

Following Errington[41] and Jain and de Pablo,[42] the L-dependent surface tension can then be calculated from the interfacial free energy as

$$\gamma_L = \frac{F_L}{2L^2}. \tag{29}$$

The surface tension can then be obtained by extrapolating the variations of γ_L with the system size to an infinitely large system $L \to \infty$, according to the following equation:[43]

$$\beta\gamma_L = \frac{a}{L^2} + \frac{b \ln L}{L^2} + \beta\gamma_\infty. \tag{30}$$

We present in Fig. 10 the results obtained in this work. Figure 10(a) shows the variation of γ_L, for $L = 8\sigma$, as a function of the temperature, while Fig. 10(b) show the dependence of γ_L on the size of the system at a reduced temperature of 0.85. Figure 10(b) also provides a comparison between our results and those obtained using the TMMC approach[41] for a cutoff radius of 3σ. Once again, the two methods lead to results in excellent agreement. Applying Eq. (30) to the TMMC results gives an estimate for γ_∞ of 0.986, while the method presented in this work gives an estimate for γ_∞ of 0.989 (fitting errors are of the order of 0.008).

B. Carbon dioxide

We present in Fig. 11 the reduced canonical partition function $Q^*(N, V, T)$ as a function of the number of CO_2 molecules for temperatures from 220 K to 280 K. The grand-canonical partition function $\Theta(\mu, V, T)$ can then be evaluated through Eq. (14). From there, similar to the case of argon discussed above, we use the formalism of statistical mechanics to calculate the thermodynamic properties of single-phase (bulk) systems and at the vapor-liquid phase boundary. To assess the accuracy of the grand-canonical partition function obtained for CO_2, we compare our results to experimental data and available simulation results.

1. Thermodynamic properties at the liquid-vapor coexistence

We start by locating the phase transition using the same methods as for argon. For each temperature, we determine

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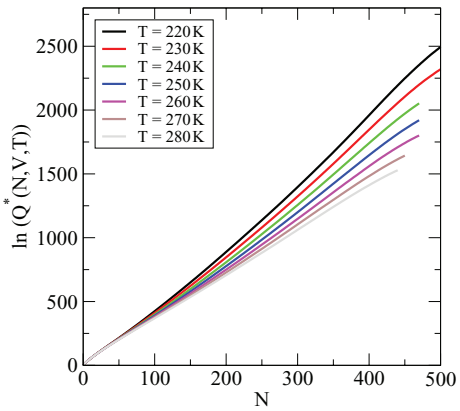

FIG. 11. Carbon dioxide: Logarithm of the reduced canonical partition function $Q^*(N,V,T) = \frac{\Lambda^{3N} Q(N,V,T)}{(\frac{8\pi^2 I k_B T}{2h^2})^N}$ against the number of CO_2 molecules.

μ_{coex}, the chemical potential at coexistence, through Eq. (21) and then P_{sat}, the saturation pressure, through Eq. (20). Figure 12 shows the variation of the saturation pressure as a function of temperature. The results obtained in this work for the saturation pressure are in excellent agreement with the experimental data. The critical pressure and temperature, extrapolated from the simulation data, are 73.51 bar and 303.78 K. Both results are in excellent agreement with the experiment (T_c = 304.19 K and P_c = 73.82 bar) and with prior simulation work, based on the Gibbs Ensemble Monte Carlo (GEMC) method[36] (T_c = 306.2 K and P_c = 77.7 bar). Applying Eqs. (22) and (23), we obtain the densities for the two coexisting phases. A comparison of the results so obtained with the experimental data is provided in Table VI. We also plot in Fig. 13 the phase diagram of CO_2 in the volume-temperature plane. Again, we obtain an excellent agreement with the experimental data. In addition, the critical volume obtained in this work (2.128 dm^3/kg) is very close to the experimental data (2.136 dm^3/kg) and to that reported in GEMC simulations[36] (2.151 dm^3/kg).

We now use the partition function obtained from the expanded Wang Landau simulations to evaluate the state

TABLE VI. Carbon dioxide: Densities at coexistence. Experimental data are taken from Ref. 38.

T(K)	ρ_{liq} (g/cm^3) this work	ρ_{liq} (g/cm^3) exp.	ρ_{vap} (g/cm^3) this work	ρ_{vap} (g/cm^3) exp.
280.00	0.893	0.883	0.116	0.121
270.00	0.943	0.944	0.086	0.089
260.00	1.002	0.998	0.062	0.064
250.00	1.041	1.046	0.044	0.046
240.00	1.083	1.089	0.032	0.033
230.00	1.123	1.130	0.022	0.021
220.00	1.156	1.167	0.015	0.016

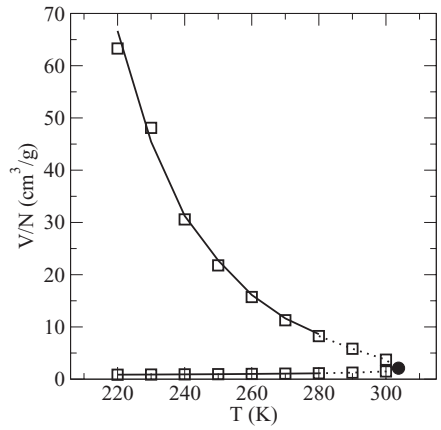

FIG. 13. Carbon dioxide: Vapor-liquid phase diagram in the volume-temperature plane. Results obtained in this work are connected by a solid line, while the experimental data are shown with squares. The critical point estimated from the simulations is shown as a filled circle.

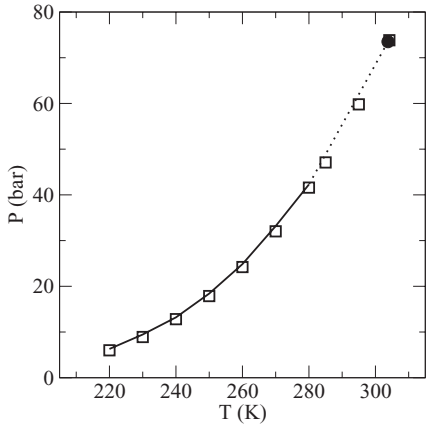

FIG. 12. Carbon dioxide: Vapor-liquid phase diagram in the pressure-temperature plane. Results obtained in this work are connected by a solid line, while the experimental data are shown with squares. The critical point estimated from the simulations is shown as a filled circle.

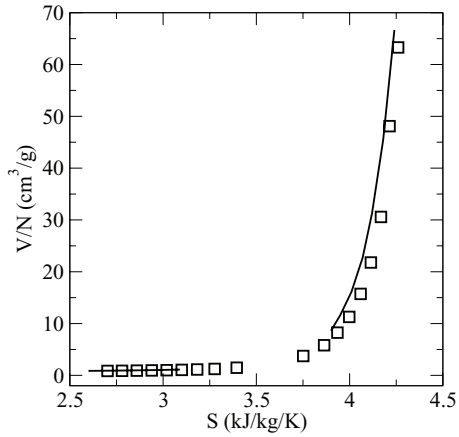

FIG. 14. Carbon dioxide: Vapor-liquid phase diagram in the volume-entropy plane. Results obtained in this work are connected by a solid line, while the experimental data are shown with squares.

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TABLE VII. Carbon dioxide: Entropies at coexistence. Experimental data are taken from Ref. 38.

T(K)	S_{liq} (kJ/kg/K) this work	S_{liq} (kJ/kg/K) exp.	S_{vap} (kJ/kg/K) this work	S_{vap} (kJ/kg/K) exp.	ΔS (kJ/kg/K) this work	ΔS (kJ/kg/K) exp.
280.00	3.12	3.18	3.90	3.93	0.78	0.75
270.00	3.04	3.10	3.95	4.00	0.91	0.90
260.00	2.95	3.02	4.01	4.06	1.06	1.04
250.00	2.89	2.94	4.07	4.11	1.18	1.18
240.00	2.82	2.86	4.12	4.17	1.30	1.31
230.00	2.74	2.78	4.18	4.21	1.40	1.43
220.00	2.64	2.70	4.24	4.26	1.60	1.56

functions at coexistence. Starting with the entropies S_{liq} and S_{vap} at coexistence

$$S_{liq} = \frac{k_B \ln \Theta(\mu,V,T)}{\langle N_{liq}\rangle} + \frac{(U_{liq}-\mu)}{k_B T},$$
$$S_{vap} = \frac{k_B \ln \Theta(\mu,V,T)}{\langle N_{vap}\rangle} + \frac{(U_{vap}-\mu)}{k_B T} \quad (31)$$

with the internal energy for the liquid U_{liq} and for the vapor U_{vap} at coexistence, Eq. (25) becomes for CO_2,

$$U_{liq} = \frac{\sum_{N>N_b}\left(E_{pot}(N)+\frac{5}{2}k_B T\right)p(N)}{\sum_{N>N_b} p(N)},$$
$$U_{vap} = \frac{\sum_{N<N_b}\left(E_{pot}(N)+\frac{5}{2}k_B T\right)p(N)}{\sum_{N<N_b} p(N)} \quad (32)$$

in which $E_{pot}(N)$ is the average potential energy per CO_2 molecule, collected during the expanded Wang Landau simulation, for a system containing N full CO_2 molecules. In this expression, $\frac{5}{2}k_B T$ is the kinetic (ideal gas) contribution arising from the 3 translational degrees of freedom and the 2 rotational degrees of freedom per CO_2 molecule.

We plot in Fig. 14 the entropies of the two coexisting phases in the volume-entropy plane and report the data in Table VII, together with the experimental data. Once again, we obtain an excellent agreement with the experiment for the liquid (with deviations of around 3%) and for the vapor (with deviations of around 1%) over the whole range of temperatures.

The Helmholtz free energies so obtained are given in Table VIII. Table VIII also provides a comparison with the corresponding experimental values. The agreement between the predicted value and the experimental data is very good for both phases, resulting in errors of less than 1.5 kJ/kg for the Helmholtz free energy of vaporization. We finally compare the results obtained for the Gibbs free energy at coexistence to the experimental data.[38] Table IX shows that the method presented here leads to Gibbs free energies in excellent agreement with the data, with a maximum deviation of about 0.5% for the temperature closest to the critical temperature and a near perfect agreement at low temperatures.

2. Thermodynamic properties of bulk phases

As for argon, the thermodynamic properties for bulk phases can be easily determined from the grand-canonical partition function. This is done by gradually varying the chemical potential with $\mu \le \mu_{coex}$ to determine the thermodynamic properties for the vapor phase and $\mu \ge \mu_{coex}$ for the liquid phase. For CO_2, the equations giving the density and the thermodynamic functions become

$$\langle\rho\rangle = \frac{\sum_N \frac{N}{V} Q(N,V,T)\exp(\beta\mu N)}{\sum_N Q(N,V,T)\exp(\beta\mu N)},$$
$$U = \frac{\sum_N\left(E_{pot}(N)+\frac{5}{2}k_B T\right)p(N)}{\sum_N p(N)},$$
$$S = \frac{k_B \ln \Theta(\mu,V,T)}{\langle N\rangle} + \frac{(U-\mu)}{k_B T},$$
$$A = U - TS. \quad (33)$$

As an example, we show in Fig. 15 the variations of density with pressure (see Fig. 15(a)) and of entropy with pressure (see Fig. 15(b)) at a temperature of 280 K. In line with the results obtained at coexistence, the density and the entropy for the bulk are in very good agreement with the experimental data.

TABLE VIII. Carbon dioxide: Helmholtz free energies at coexistence. Experimental data are taken from Ref. 38.

T(K)	A_{liq} (kJ/kg) this work	A_{liq} (kJ/kg) exp.	A_{vap} (kJ/kg) this work	A_{vap} (kJ/kg) exp.	ΔA(kJ/kg) this work	ΔA(kJ/kg) exp.
280.00	−967.69	−970.8	−997.71	−999.8	−30.02	−29.0
270.00	−934.83	−940.0	−968.00	−973.3	−33.17	−33.3
260.00	−904.66	−909.5	−940.95	−945.1	−36.29	−35.6
250.00	−875.54	−879.3	−915.05	−915.3	−39.51	−36.0
240.00	−848.73	−850.1	−887.32	−888.3	−38.59	−38.2
230.00	−819.12	−821.7	−861.24	−862.7	−42.12	−41.0
220.00	−794.1	−800.7	−834.39	−839.6	−40.29	−38.7

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TABLE IX. Gibbs free energy for CO_2 at coexistence. Experimental data are taken from Ref. 38.

T(K)	280	270	260	250	240	230	220
G (kJ/kg) (this work)	−960.53	−931.00	−902.10	−873.88	−846.50	−819.83	−793.41
G (kJ/kg) (exp)	−966.88	−936.6	−906.84	−877.35	−848.66	−820.79	−794.4

3. Surface tension calculations

The surface tension of CO_2 can also be determined from the simulations presented in this work. This can be done according to the same process as that outlined for argon (see Eqs. (28) and (29)). Based on the system sizes studied in this work, we obtain an estimate for the surface tension for an infinitely large system of 14 ± 1 mN.m^{-1} at $T = 238$ K, in reasonable agreement with the value of 12 mN.m^{-1} for the experimental value[44] and obtained on another potential model for carbon dioxide.[45]

V. CONCLUSION

In this work, we show how we use the Wang-Landau method to obtain a high-accuracy estimate for the grand-canonical partition function of atomic and molecular fluids. Prior applications to the calculation of the thermodynamic properties of realistic fluids had been limited, partly because of a lack of accuracy of the partition function so obtained. By combining the WL sampling with a simplified version of the expanded grand-canonical ensemble, we show how the grand-canonical partition function of the system can be evaluated. Then, using the formalism of statistical thermodynamics, thermodynamic properties such as the entropy and the Helmholtz and Gibbs free energies are determined for argon and carbon dioxide over a wide range of conditions. We establish the accuracy of the method by comparing our results to prior simulation work, when available, and to experimental data in the bulk phases and at coexistence. We emphasize that the method presented in this work is general and can be applied to any kind of molecular fluid. This method is also simple to use since a single simulation run yields the thermodynamic properties of the system at a given temperature.

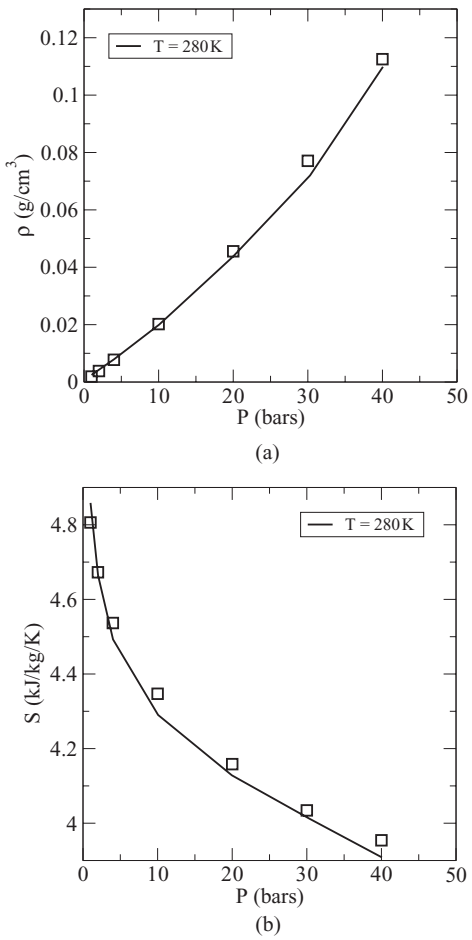

FIG. 15. Single-phase bulk properties for carbon dioxide: Variations of (a) density and (b) entropy as a function of pressure for the vapor phase. Results obtained in this work are shown with lines, while the experimental data are shown as squares.

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